

Basic Linear Algebra (융합기초선형대수학)

KNU Math 902

Classnotes

Mark Siggers

These notes are supplemental to a class taught from Trefethen and Bau's 'Numerical Linear Algebra'. It is expected that students have this text. It is herein referred to as 'the text'.

v. September 3, 2026

The text is on Numerical Linear Algebra, and supposes a familiarity with classical linear algebra. It expects that you know what matrices and vectors are, how matrix equations are a way of writing systems of linear equations, and how we can solve them using Gaussian Elimination. You should be familiar eigenvalues and eigenvectors of a matrix, have some notion that they are important, and have seen how to find them algebraically, even if you don't remember this completely.

It doesn't matter so much if you can remember it, because in practice, we aren't going to use Gaussian Elimination. In practice, we are using big matrices, and solving them with computers, so are generally not solving them algebraically.

Gaussian Elimination is a great theoretical tool for understanding the theory of matrices, for diagonalising matrices or finding their nullspace. We will 'recall' such things, but only briefly.

We will spend much more of our time on matrix decompositions and approximations. We will look first at the QR-decomposition of a matrix, and because matrices are seldom square, we will talk more of the single value decomposition than diagonalisation.

Our focus will be on not the simplest algorithms for working with matrices, but on those that work best on computers— stable algorithms, which control their error. It is a different game than classical linear algebra, but it has its own beauty.

Lecture 1 Matrix Vector Multiplication

In this book, we work over the complex numbers $\mathbb{C}$, but it is usually okay to think of working over the reals $\mathbb{R}$. Probably all pictures, and much intuition, will be of the real case.

An $m \times n$ matrix $A = A_{m \times n}$ over the complex numbers $\mathbb{C}$ is a ‘rectangle’ of m rows and n columns of numbers from $\mathbb{C}$.

$$A = \begin{bmatrix} 1 & 2 & \frac{1}{2} \\ 4 & i & \pi \end{bmatrix}$$

is a 2×3 matrix. We write $A = [a_{i,j}]$ to designate that $a_{i,j}$ is the entry in the i^{th} row (from the top) and the j^{th} column. We drop the comma and write a_{ij} when it does not cause confusion. The values m and n are the *dimensions* of the matrix. A $1 \times n$ matrix is a (*column*) *vector* and a $m \times 1$ matrix is a (*row*) *vector*. When $A = [a_{ij}]$ we write a_{i*} for the i^{th} row vector and a_{*j} for the j^{th} column vector. The ‘ $*$ ’ runs over all values.

Vectors in $\mathbb{C}^m$ are viewed sometimes as points in $\mathbb{C}^m$, and sometimes as the ‘arrow’ from the origin $0 = (0, 0, \dots, 0)$ to the point.

Practice

Where $A = [a_{ij}]$ is the 2×3 matrix $A = \begin{bmatrix} 1 & 2 & \frac{1}{2} \\ 4 & 0 & \pi \end{bmatrix}$, what is a_{23} ? What is a_{*2} ?

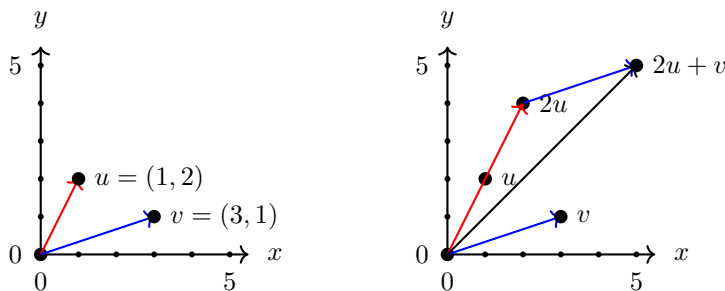
We add matrices of the same dimensions componentwise

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 4 \end{bmatrix}$$

and multiply matrices by *scalars* $r \in \mathbb{R}$ componentwise

$$2 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 4 & 6 \end{bmatrix}.$$

For vectors there is an obvious interpretation of this.



We can also multiply matrices, but only if their dimensions are right. Where $A = A_{\ell \times m} = [a_{ij}]$ and $B = B_{m \times n} = [b_{jk}]$ the product $C_{\ell \times n} = AB$ is the matrix $C = [c_{ik}]$ where

$$c_{ik} = \sum_{j=1}^m a_{ij}b_{jk} = a_{i1}b_{1k} + a_{i2}b_{2k} + \cdots + a_{im}b_{mk} = a_{i*} \cdot b_{*k}.$$

The entry $c_{ik} = a_{i*} \cdot b_{*k}$ is the dot product of the i^{th} row of A and the k^{th} column of B :

$$\begin{bmatrix} \cdot & \cdot & \cdot \\ \boxed{1} & \boxed{4} & \boxed{2} \\ \cdot & \cdot & \cdot \end{bmatrix} \begin{bmatrix} \cdot & \boxed{3} & \cdot & \cdot \\ \cdot & \boxed{1} & \cdot & \cdot \\ \cdot & \boxed{1/2} & \cdot & \cdot \end{bmatrix} = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 8 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}.$$

In the *row view of the matrix product* AB , ‘the rows of A are coefficients in linear combinations of the rows of B ’:

$$\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 & 7 \\ 2 & 8 \end{bmatrix} = \begin{bmatrix} 1 \cdot \begin{bmatrix} 3 & 7 \end{bmatrix} + 4 \cdot \begin{bmatrix} 2 & 8 \end{bmatrix} \\ 2 \cdot \begin{bmatrix} 3 & 7 \end{bmatrix} + 3 \cdot \begin{bmatrix} 2 & 8 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 11 & 39 \\ 12 & 38 \end{bmatrix}.$$

In the *column view* we view the columns of B as providing coefficients in linear combinations of the columns of A .

$$\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 & 7 \\ 2 & 8 \end{bmatrix} = \begin{bmatrix} \begin{matrix} 3 \\ \times \\ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \end{matrix} & \begin{matrix} 7 \\ \times \\ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \end{matrix} \\ \begin{matrix} 2 \\ + \\ 2 \end{matrix} & \begin{matrix} 8 \\ + \\ 8 \end{matrix} \\ \begin{matrix} \times \\ \begin{bmatrix} 4 \\ 3 \end{bmatrix} \end{matrix} & \begin{matrix} \times \\ \begin{bmatrix} 4 \\ 3 \end{bmatrix} \end{matrix} \end{bmatrix} = \begin{bmatrix} 11 & 39 \\ 12 & 38 \end{bmatrix}$$

Practice

Compute the two products

$$\begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} \begin{bmatrix} a & b & c \end{bmatrix}$$

and observe that **products do not commute**– the order of the matrices is important when we are multiplying them. These are two different products of two vectors. The first is called the *inner product* or the *dot product*, and the second is called the *outer product*.

Practice

Find the 3×3 matrix A such that

$$\begin{bmatrix} 3 & 2 & 0 \\ 1 & 2 & 2 \\ 4 & 2 & 1 \end{bmatrix} = A \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Matrix times a vector

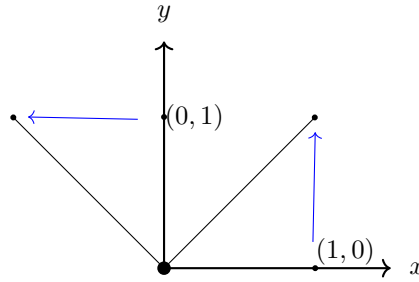
When the second matrix is a column vector $x \in \mathbb{C}^n$, multiplication

$$Ax = b$$

has two interpretations.

One interpretation is to view A as transforming the vector $x \in \mathbb{C}^n$ to a vector b in $\mathbb{C}^m$.

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$



It is a linear transformation:

$$A(x + y) = Ax + Ay \quad \text{and} \quad A(\alpha x) = \alpha Ax.$$

Every linear transformation from $\mathbb{C}^n$ to $\mathbb{C}^m$ can be described as multiplication by an $m \times n$ complex matrix A .

The other interpretation views b as the linear combination of columns of A with co-efficients given by x :

$$\begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 2 \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 1 \cdot \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}.$$

By the column view of the product $Ax = b$ we view b as a vector generated by x as a linear combination of columns of A .

Range and Nullspace of A

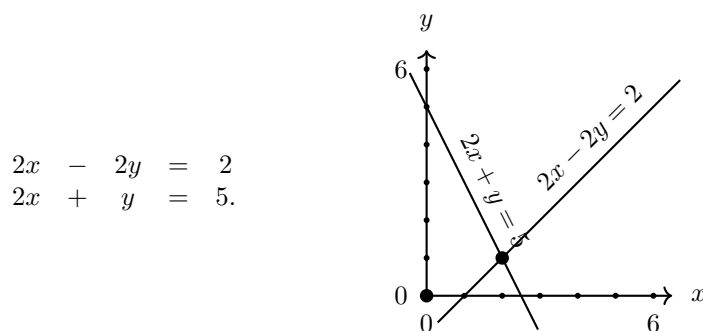
The *range*, $\text{range}(A)$, of an $m \times n$ matrix A is the set of vectors in $\mathbb{C}^n$ that can be generated as linear combinations of its columns. It is clearly a subspace of $\mathbb{C}^n$ so more commonly called the *column space* of A . The *nullspace*, $\text{null}(A)$, of A is the set of vectors x in $\mathbb{C}^m$ such that $Ax = 0$. This is also clearly a space, as $Ax = 0$ and $Ay = 0$ implies that

$$A(ax + by) = aAx + bAy = a0 + b0 = 0.$$

The range is the set of values b such that the equation $Ax = b$ has a solution and the nullspace is the set of values x such that $Ax = 0$ has a solutions. This is not in the text, but lets recall how we solve these equations in a traditional linear algebra course.

The equation $Ax = b$ and Gaussian Elimination

Recall that a system of linear equations, each row of which can be viewed as a hyperplane in real space,



can be written as a matrix equation of the form $Ax = b$:

$$\begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}.$$

In highschool you solved a system of equations like this. Your first step was to ‘take a copy of the first equation from the second,’ to get an equivalent system of equations.

$$\begin{array}{rclcl} 2x & - & 2y & = & 2 \\ 2x & + & y & = & 5 \end{array} \rightsquigarrow \begin{array}{rclcl} 2x & - & 2y & = & 2 \\ & & 3y & = & 3. \end{array}$$

The second system is the matrix equation

$$\begin{bmatrix} 2 & -2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

You should understand how to solve matrix equations in this way, using ‘row operations’ to reduce this to something like

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

from which you can easily read off the solution $(x, y) = (2, 1)$. This is called Gaussian elimination. You might have seen it done using *augmented matrices* to reduce your work:

$$\left[\begin{array}{cc|c} 2 & -2 & 2 \\ 2 & 1 & 5 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 2 & -2 & 2 \\ 0 & 3 & 3 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 1 & 1 \end{array} \right] \rightsquigarrow \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array} \right].$$

Sometimes things didn’t work so nicely. Sometimes there was no solution, and sometimes there were many. We looked at rank and determinants and inverses to characterise when $Ax = b$ has a solution.

Rank

The *rank* of a matrix is the number independent columns it has; equivalently, and we will see this later, the number of independent rows. By this definition, the rank of an $m \times n$ matrix A is at most the minimum of m and n . If it is exactly $\min(m, n)$ it is *full rank*. A square $m \times m$ matrix A has full rank $\text{rank}(A) = m$ if and only if the following equivalent conditions are true.

- i. Gaussian elimination reduces A to the identity matrix I .
- ii. A has an inverse matrix A^{-1} such that $AA^{-1} = I = A^{-1}A$.
- iii. $\text{range}(A) = \mathbb{C}^m$.
- iv. $\text{null}(A) = \{0\}$.
- v. $\det(A) \neq 0$.
- vi. A is non-singular.
- vii. 0 is not an eigenvalue of A .
- viii. The columns of A are linearly independent.
- ix. The rows of A are linearly independent.

Recall that the determinant $\det(A)$ is the product of the diagonals we get after doing Gaussian elimination on A without row switches or scaling; we don’t talk about it much in this class.

A square matrix is *singular* if it does not have full rank, and non-singular if it does; think of this as a definition.

A number λ is an *eigenvalue* of A if there is a non-zero solution to $Ax = \lambda x$. These we will talk about.

A set $V = \{v_1, \dots, v_d\}$ of vectors in $\mathbb{C}^m$ is *linearly independent* if no one vector can be expressed as a linear combination of the others:

$$v_d = x_1 v_1 + x_2 v_2 + \dots + x_{d-1} v_{d-1};$$

equivalently, if

$$([v_1 \mid v_2 \mid \dots \mid v_d] x = 0) \rightarrow x = 0.$$

A matrix inverse times a vector

If we have the inverse A^{-1} of A , we can solve the equation $Ax = b$ by computing $x = A^{-1}b$. But we are usually not going to do this. Finding A^{-1} will often be computationally difficult. So the text urges you not to view $x = A^{-1}b$ as a transformation of b . They suggest that it is better viewed as a change of basis. When $A = [a_1 | a_2 | \dots | a_n]$ has independent columns a_i , $x = (x_1, \dots, x_n)$ is the vector of co-efficients x_i such that

$$b = x_1 a_1 + x_2 a_2 + \dots + x_n a_n.$$

We view $x = A^{-1}b$ as these co-efficients of b expressed in a basis of $\text{range}(A)$.

Problems from the text

Lecture 1: 1, 3, 4. (I can't understand # 2. Can anyone explain it?)

Extra for the exercises

When we were doing Gaussian elimination above we reduced

$$\left[\begin{array}{cc|c} 2 & -2 & 2 \\ 2 & 1 & 5 \end{array} \right] \quad \text{to} \quad \left[\begin{array}{cc|c} 2 & -2 & 2 \\ 0 & 3 & 3 \end{array} \right].$$

This was a row operation, we removed a copy of the first row from the second. We can view this as multiplying the first matrix, on the left, by the elementary matrix $E = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$. Under the row view of matrix multiplication, the first row of E says 'do nothing to the first row' and the second row of it says 'remove one copy of the first row from the second row':

$$\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \left[\begin{array}{cc|c} 2 & -2 & 2 \\ 2 & 1 & 5 \end{array} \right] = \left[\begin{array}{cc|c} 2 & -2 & 2 \\ 0 & 3 & 3 \end{array} \right].$$

Exercise 1 is asking you to start with a matrix B and do row and column operations. For the row operations, you will multiply by some matrix on the left, for the column operations, you will do so on the right. Be careful of the order.

Lecture 2 Orthogonal Vectors and Matrices

The *transpose* of a matrix $A = [a_{ij}]$ is the matrix $A^T = [a_{ji}]$ that we get by reflecting it through its diagonal. A matrix A is *symmetric* if $A = A^T$.

Recall that a complex number is of the form

$$z = a + ib$$

where $a, b \in \mathbb{R}$ and i is the positive square root of -1 . The *adjoint* of a complex number $z = a + ib$ is $\bar{z} = a - ib$. More generally the *adjoint* of a matrix $A = [a_{ij}]$ is the matrix

$$A^* = [\bar{a}_{ji}]$$

that we get from it by taking its transpose, and replacing each entry with its adjoint. If $A^* = A$ then A is *hermitian* and a real hermitian matrix is *symmetric*.

Practice

For complex numbers a and b show that $\overline{ab} = \bar{a}\bar{b} = \bar{b}\bar{a}$. Using this show for complex matrices A and B that $(AB)^* = B^*A^*$.

Most (with some definite exceptions) of the theory of adjoints and hermitian matrices goes through without adjustment to transposes and symmetric matrices when we restrict to real matrices; so many books will use A^T much more than A^* , and often use A^H for A^* . Don't be surprised if I use T occasionally, instead of $*$.

Inner product

The *inner product* of two vectors a and b in $\mathbb{C}^m$ is the product a^*b where they are viewed as column vectors. This is also called the *dot product* of the vectors, so again, don't be surprised if I write it $a \cdot b$.

Practice

Compute a^*b when $a = (2, 3, 4)$ and $b = (3, 1, -1)$. When $a = (-i, 2, 1)$ and $b = (i, i, -2)$.

A vector $v = (v_1, \dots, v_n)$ can be viewed as a point in $\mathbb{C}^n$, but also it can be viewed as the directed distance from 0 to that point. As such, its distance can be computed by the Pythagorean Theorem as

$$\|v\| = \sqrt{v^*v}.$$

A vector is *normal* if $\|v\| = 1$.

Practice

Compute $\|(3, 4)\|$.

Using the Law of Cosines, one can show that where α is the angle between vectors a and b (viewed as arrows from the origin 0), that

$$\cos \alpha = \frac{a^*b}{\|a\|\|b\|}.$$

This is 0 for non-zero a and b if and only if they are perpendicular.

Orthogonal Vectors

Non-zero vectors a and b such that $a^*b = 0$ are called *orthogonal*. Sets $X, Y \subset \mathbb{C}^m$ of non-zero vectors *orthogonal* if $x^*y = 0$ for each $x \in X$ and $y \in Y$. A set Q of non-zero vectors is *orthogonal* if $x^*y = 0$ for any two vectors $x, y \in Q$; if the vectors are also normal, then Q is *orthonormal*.

When solving $Ax = b$ we are looking to express the vector b as a linear combination of the columns of A . This is especially easy when the columns of A are a set of orthonormal vectors A .

Indeed assume that $A = [q_1 \mid \cdots \mid q_d]$ where the set $\{q_1, \dots, q_d\}$ is orthonormal. If $Ax = b$ for $x = (x_1, \dots, x_d)$, then we have that

$$b = x_1q_1 + \dots + x_dq_d$$

and we can find x_i by computing

$$q_i^*b = q_i^*(x_1q_1 + \dots + x_dq_d) = x_1q_i^*q_1 + \dots + x_iq_i^*q_i + \dots + x_dq_i^*q_d = x_i.$$

In particular, if there is a solution to $Ax = b$ then it is unique:

$$x = \sum_{i=1}^d (q_i^*b)q_i = \sum_{i=1}^d (q_iq_i^*)b.$$

The uniqueness gives us the following.

Fact 2.1. *A set of orthogonal vectors is independent.*

The two expressions $(q_i^*b)q_i$ and $(q_iq_i^*)b$ are the same but have different interpretations. In the first, q_i^*b is a scalar, it is the co-efficient of q_i in the expansion of b . In the second $(q_iq_i^*)$ is a matrix that projects b onto its i^{th} component.

Even if there is no solution to $Ax = b$, we can write b as

$$b = x_1q_1 + \dots + x_dq_d + r$$

for some r that is orthogonal to the range of A .

Unitary Matrices

A square matrix U is *unitary* (or *orthogonal* in the real case) if its columns are orthonormal. In this case $U^*U = I$ and so $U^{-1} = U^*$. As a transformation, a unitary matrix is quite simple. It preserves the lengths of vectors and the angle between them, so is a rigid rotations of $\mathbb{C}^m$. Indeed

$$(Qu)^*(Qv) = u^*Q^*Qv = u^*Iv = u^*v$$

gives that $\|Qu\| = (Qu)^*(Qu) = u^*u = \|u\|$ and that the cosine of the angle between Qu and Qv is

$$\frac{(Qu)^*(Qv)}{\|Qu\|\|Qv\|} = (Qu)^*(Qv) = u^*v = \frac{u^*v}{\|u\|\|v\|}$$

which is the cosine of the angle between u and v .

Problems from the text

Lecture 2: 1-6. (1-4 are classical linear algebra theory.)

Lecture 3 Norms

In a classical linear algebra course we usually only consider one vector norm, the euclidean norm, because, in a real sense, all are equivalent. But numerically, other norms are often easier to work with. And when used to define matrix norms, something that is not always seen in a classical class, they have some interesting applications.

A norm is a non-negative function on vectors that has many properties similar to what you would expect of a 'length' function. Technically, a (*vector*) *norm* is a non-negative function $\|\cdot\| : \mathbb{C}^n \rightarrow \mathbb{R}^+$ that satisfies, for all $c \in \mathbb{C}$ and $v, w \in \mathbb{C}^n$,

- $\|v\| = 0$ if and only if $v = 0$, (positive definiteness),
- $\|c \cdot v\| = |c| \cdot \|v\|$, (scaling), and
- $\|v + w\| \leq \|v\| + \|w\|$, (the triangle inequality).

For any $p \geq 1$ the function

$$\|v\|_p = \left(\sum_{i=1}^n |v_i|^p \right)^{1/p}$$

is a norm. The most common cases we consider are when $p = 1$ or 2 , or one defined by taking $p \rightarrow \infty$:

- the ℓ^1 - or *1-norm* $\|v\|_1 = |v_1| + |v_2| + \dots + |v_n|$,
- the ℓ^2 - or *Euclidean norm* $\|v\| = \|v\|_2 = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$,
- the ℓ^∞ - or *max norm* $\|v\|_\infty = \max\{|v_1|, |v_2|, \dots, |v_n|\}$.

Problem:

Show that $\|\cdot\|_1$, $\|\cdot\|_2$ and $\|\cdot\|_\infty$ are norms.

Problem:

For each of the above norms $\| \cdot \|$, draw the unit sphere $S = \{v : \|v\| = 1\}$ in $\mathbb{R}^2$.

A common problem is to minimise or maximize a norm on the set of vectors in a subspace.

Problem:

Minimise the above norms on the vectors in the line $L : x/3 + y/4 = 1$.

When you do this, you will notice that the minimum point for the ℓ_1 -norm is mostly 0s in such problems. While the other norms tend to have non-zero numbers in more coordinates. Because of this the ℓ_1 -norm will often be a useful measure in approximations.

Apart from the p -norms, we also use weighted norms. For a (usually diagonal) non-singular matrix W let $\|v\|_W = \|Wv\|_2$.

Problem:

When $W = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ draw the unit sphere S for the norm $\|\cdot\|_W$.

General matrix norms

We also define norms $\| \cdot \|$ for matrices. We want that they have the same properties of positive definiteness, scaling, and triangle inequality that vector norms have, but we want that they also have sub-multiplicativity:

$$\|AB\| \leq \|A\|\|B\|.$$

Viewing an $m \times n$ matrix $A = [a_{ij}]$ as a vector of length mn :

$$(a_{11}, \dots, a_{1n}, a_{21}, \dots, a_{2n}, \dots, a_{m1}, \dots, a_{mn})$$

we immediatly get the first three properties, but we don't always get sub-multiplicativity. In the case of the 2-norm though, we do get multiplicativity by Cauchy-Schwarz, and so we do use this construction for the *Frobenius Norm*. Where $A = [a_{ij}]$ we define

$$\|A\|_F = \|(a_{11}, \dots, a_{1n}, a_{21}, \dots, a_{2n}, \dots, a_{m1}, \dots, a_{mn})\|_2.$$

Problem

Show that the Frobenius norm satisfies sub-multiplicativity.

Matrix norms induced by vector norms

While other vector norms don't give matrix norms in this way, they do give them in another way.

For any vector norm $\|\cdot\|$ let

$$\|A\| = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \max_{x \in S} \|Ax\|$$

be the maximum stretch of x by A in this norm. Here S is the unit sphere with respect to the given norm.

Problem:

Show that submultiplicativity is satisfied above for any choice of the vector norm $\|\cdot\|$.

To compute the norm of a matrix A induced by a vector norm, we only have to look at all vectors in the unit sphere S under that vector norm, and check which one is stretched the most by A .

In the text they draw the action of a matrix A on the unit balls under the various vector norms, and show how one can visualise the induced matrix norms from these pictures.

As for the Frobenius norm, computing the 1-norm and the ∞ -norm of a matrix are easy.

Example 3.1. To compute the 1-norm $\|A\|_1$ we for a matrix $A = [a_1 \mid \cdots \mid a_n]$ we maximise

$$\|Ax\| = x_1|a_1| + \cdots + x_n|a_n|$$

over x such that $x_1 + x_2 + \cdots + x_n = 1$. This is clearly maximised when x is the standard normal vector e_M with a 1 in the M^{th} coordinate where a_M is the largest column. So $\|A\|_1 = |Ae_M| = |a_M|$. The 1-norm of a matrix A is the maximum of the norms of its columns.

With similar analysis one can do the following.

Problem

Characterise the vectors in the unit sphere under the vector norm $\|\cdot\|_\infty$. Show that for a matrix A $\|A\|_\infty$ is the max, over all rows of A , of the sum of the entries in the row.

For other norms it is a bit harder computationally, though visually it is not so bad.

Example 3.2. Under the 2-norm, the unit sphere S is our usual unit sphere. Applying A to transform each point x in S to Ax transforms S into an ellipsoid. The matrix norm $\|A\|_2$ is the length of the longest axis of this ellipsoid.

Problems from the text

Lecture 3: 1-6